

HW 6: Analysis and Limit F-signature

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Problem 1

Prove that $\lim_{n \rightarrow \infty} \frac{1}{n^3} = 0$.

Proof. Let $\epsilon > 0$ be arbitrary, then we choose an $N > \frac{1}{\sqrt[3]{\epsilon}}$ such that $\forall n > N$ we have

$$\left| \frac{1}{n^3} \right| < \left| \frac{1}{\left(\frac{1}{\sqrt[3]{\epsilon}}\right)^3} \right| = \epsilon.$$

Thus, the sequence converges to 0 as n approaches infinity. $\square$

Problem 2

Let $f_n(x) = \frac{1}{x^{2n+1}}$ be a sequence of functions, viewing $f_n : [1, \infty) \rightarrow \mathbb{R}$. Prove that $\lim_{n \rightarrow \infty} f_n(x)$ exists.

Proof. We follow with a proof by cases.

Case 1. $x > 1$

For all $\epsilon > 0$ there exists some $N > -\left(\frac{\ln \epsilon}{2 \ln x} + \frac{1}{2}\right)$, then $\forall n > N$, we have

$$\left| \frac{1}{x^{2n+1}} \right| < \left| \frac{1}{x^{2\left(\frac{\ln \epsilon}{2 \ln x} - \frac{1}{2}\right)+1}} \right| = x^{\frac{\ln \epsilon}{\ln x}} = \epsilon.$$

Thus, when $x > 1$, $\lim_{n \rightarrow \infty} f_n(x) = 0$.

Case 2. $x = 1$

For all $\epsilon > 0$ there exists some $N > -\left(\frac{\ln(\epsilon+1)}{2 \ln x} + \frac{1}{2}\right)$ where $\forall n > N$, we have

$$\left| \frac{1}{x^{2n+1}} - 1 \right| < \left| \frac{1}{x^{-2\left(\frac{\ln(\epsilon+1)}{2 \ln x} + \frac{1}{2}\right)+1}} - 1 \right| = x^{\frac{\ln(\epsilon+1)}{\ln x}} - 1 = \epsilon + 1 - 1 = \epsilon.$$

Thus, when $x = 1$, $\lim_{n \rightarrow \infty} f_n(x) = 1$. $\square$

Problem 3

Part a.

- (i) 15
- (ii) 22
- (iii) 25

Part b.

I am going to omit the instructions and jump into the proof, as the formatting is a little tedious!

Proof. Let us recall that

$$\dim_{\mathbb{F}_p} \left(\frac{\mathbb{F}_p[x_1, \dots, x_n]}{\langle x_1^p, \dots, x_n^p \rangle} \right) = p^n.$$

We now must add a generator f^k to the ideal. We follow with a proof by cases.

Case 1. For $f = x_1^{a_1} \dots x_n^{a_n}$ we have $\exists i, a_i k > p$.

Suppose there exists such a degree in f , then it follows that f is a multiple of x_i , thus

$$\dim_{\mathbb{F}_p} \left(\frac{\mathbb{F}_p[x_1, \dots, x_n]}{\langle x_1^p, \dots, x_n^p \rangle} \right) = \dim_{\mathbb{F}_p} \left(\frac{\mathbb{F}_p[x_1, \dots, x_n]}{\langle x_1^p, \dots, x_n^p, f^k \rangle} \right) = p^n.$$

Case 2. $\forall i, a_i k \leq p$

Suppose there is a degree of a variable in f^k less than p , then it follows that $f^k, x_1^p, \dots, x_n^p$ are divisible by some $x^{a_i k}$; $a_i k \leq p$, then we must count the number of monomials not in our initial ideal and in our new ideal.

Let us suppose $x^c \notin \langle x_1^p, \dots, x_n^p \rangle$ with each member of c denoted by c_i , then we must have $a_i k \leq c_i < p$, as f^k must divide x^c , which also must divide $x_1^p \dots x_n^p$. Thus, we have the number of elements in

$$\{(c_1, \dots, c_n) \in \mathbb{N}^n \mid \forall i, a_i k \leq c_i < p\}.$$

We then isolate the left hand side to get $p - a_i k$, and since this property is independent of a fixed c , we have

$$p^n - \prod_{i=1}^n (p - a_i k)$$

□

Part c.

Proof. Suppose $\frac{k}{p} \rightarrow t$, then $k \rightarrow pt$. Additionally, in the case of $a_i k \leq p$, we want a_i to be held for all i , thus we should have $t \leq \min_i \{\frac{1}{a_i}\}$. Furthermore, we are able to factor out p from

$$\prod_{i=1}^n (p - a_i k)$$

n times. Thus, we may factor p^n out of $p^n - \prod_{i=1}^n (p - a_i k)$ to get

$$1 - \prod_{i=1}^n (1 - a_i t)$$

after dividing by p^n and subtracting 1. For the other case of $a_i k > p$, we simply divide by p^n and subtract 1 to get 0. $\square$